

# CYBE 435

## Computer Incident Response

### Lecture 4:

### Detection and Identification

# Announcements

- National Cyber League
  - Many of you are actively working in the NCL Gymnasium
  - For the Gymnasium portion of your grade, I will award full credit for 2000 points by the start of the Individual Game on Friday, October 24.
- Quiz #1 is due by tonight at 11:59pm
- Incident Management Policy is due by next Tuesday, Sept. 9 at 11:59pm

# Consider the following security incident

- Cloudflare is a cloud platform that provides improved speed, reliability, and security for large web servers
- In February 2017, researchers discovered a bug that occasionally leaked small chunks of memory from unrelated web sessions
- This bug had the potential for passwords, session tokens, and encryption keys to leak out.
- Malicious actors could have used this exploit to impersonate users, hijack accounts, and access sensitive information

# Detection and Response to Cloudflare bug

- Google researchers engaged in a web crawling activity noticed unusual output
- They reported the problem to Cloudflare immediately
- Cloudflare's security team acted within hours
  - Verified the problem
  - Identified the faulty code
  - Patched the problem
- Cloudflare notified customers and worked to purge web archives
- There were no known malicious exploits from this incident

# What types of tools are available to detect incidents?

- Host and Endpoint Monitoring
  - Anti-virus or Anti-Malware software
  - File and Log Monitoring
- Network Traffic Analysis
  - Firewalls and Intrusion Detection Systems
- Security Information and Event Management (SIEM) systems
  - Automated aggregation and analysis of log data
  - Many commercial security companies provide this service
- Users and Community (for example, the Cloudflare Incident)
- We will discuss these tools more in next Lecture

# What types of Cybersecurity Incidents do we need to detect?

- Ransomware
  - Malicious software that encrypts or destroys critical data
  - High cost of recovery and significant downtime
  - Examples: WannaCry, NotPetya, Ryuk, LockBit
- Compromise of User Credentials
  - Stolen or guessed user names and passwords
  - Could be through phishing or through brute force attacks
  - Could access sensitive accounts or data
  - Many significant breaches begin with the breach of one account and escalate from there

# Types of Cybersecurity Incidents

- Phishing and Social Engineering
  - Deceptive emails, texts or calls designed to trick a user into giving up information or clicking on a link that allows installation of malware
  - Can result in transfer of money or credentials
  - Variations: spear phishing, whaling.
  - AI makes communications harder to detect
  - Many more serious incidents begin with phishing attacks
  - Most common attack and low cost for attackers

# Other Important Types of Cybersecurity Attacks

- Data breaches
  - Denial of Service
  - Supply Chain or Third Party Compromise
  - Insider Threats
  - Configuration Errors and Unpatched Systems
  - Critical Infrastructure Attacks
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- In the next lecture, we will begin to look at tools for detecting and identifying these incidents in more detail.